IBM Developer
SKILLS NETWORK

WINNING SPACE RACE WITH DATA SCIENCE

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Outline

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Executive Summary

The goal of this capstone project will allow us to determine if the SpaceX Falcon 9 first stage will land successfully using several machine learning algorithms.

- Summary of methodologies
  - The primary steps used in this project are as follows:
    - Data Collection through API and Web Scraping
    - Data Wrangling to Transform Data
    - Exploratory Data Analysis with SQL and Data Visuals
    - Predictions with Machine Learning
- Summary of all results
  - The results will be shared through a variety of formats:
    - Data Analysis
    - Data Visuals
    - Predictive Analysis
    - Interactive Analytics

Introduction

Project Background and Context

SpaceX is the most successful company of the commercial space age, making space travel affordable. The company advertises Falcon 9 rocket launches on its website, with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. Based on public information and machine learning models, we are going to predict if SpaceX will reuse the first stage.

Problems You Want To Find Answers To

- How do variables such as payload mass, launch site, number of flights, and orbits affect the success of the first stage landing?
- Does the rate of successful landings increase over the years?
- What is the best algorithm that can be used for binary classification in this case?

METHODOLOGY

SECTION 1

Methodology

Executive Summary

Data collection methodology:

- SpaceX Rest API
- Web Scraping Falcon 9 and Falcon heavy launch records from Wikipedia

Perform data wrangling

- Filter the data
- Deal with missing values
- One Hot Encoding to prepare the data to a binary classification

Perform Exploratory Data Analysis (EDA) using SQL and visualization

Perform Interactive Visual Analytics using Folium and Plotly Dash

Perform Predictive analysis using classification models

Build, tune and evaluate classification models for the best results

- Create a column for class, train/test split data, find the best algorithm using test data, standardize (and transform) the data

Data Collection

Data collection is the process of gathering data using both SpaceX API and web scraping from Wikipedia. The data can be structured, unstructured and semi-structured.

Data Columns are obtained by using:

SpaceX REST API - FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude Data Columns are obtained by using Wikipedia

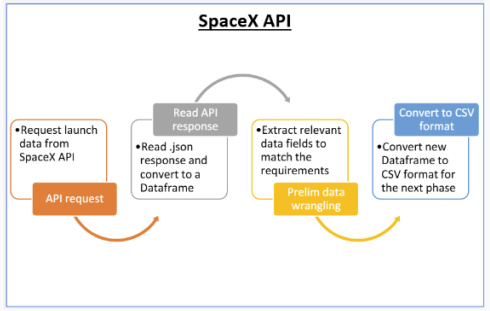
Web Scraping - Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

GitHub URL

Data Collection

SpaceX API Steps

- ❑ Request Data
- ❑ Decode Response
- ❑ Request Information
- ❑ Create Dictionary
- ❑ Create DF
- ❑ Filter DF
- ❑ Replace Missing Values
- ❑ Export Data

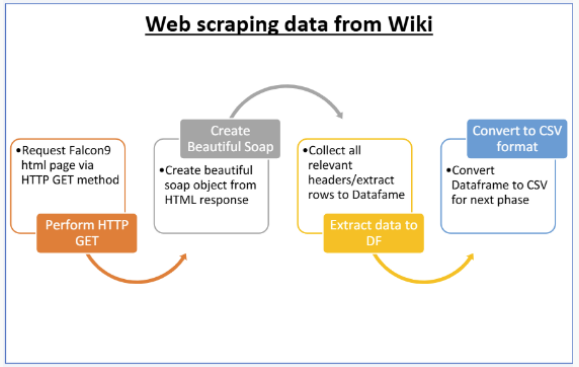


GitHub URL

Data Collection

Web Scraping Steps

- ❑ Request Data
- ❑ Create BeautifulSoup Objects
- ❑ Extract Column Names
- ❑ Collect Data
- ❑ Create Dictionary
- ❑ Create DF
- ❑ Export Data



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Data Collection

Data Wrangling Steps

There are several different cases in which the booster did not land successfully in this data set

- True Ocean = mission outcome was successfully landed to a specific region of the ocean
- False Ocean = mission outcome was unsuccessfully landed to a specific region of the ocean.
- True RTLS = mission outcome was successfully landed to a ground pad
- False RTLS = mission outcome was unsuccessfully landed to a ground pad
- True ASDS = mission outcome was successfully landed on a drone ship
- False ASDS = mission outcome was unsuccessfully landed on a drone ship.

Training Label 1 = successfully landing
Training Label 0 = unsuccessful landing

Data Collection

Data Wrangling Steps (Cont'd)

- ❑ Perform EDA and Determine Data Labels
- ❑ Calculate the Number of Launches For Each Site
- ❑ Calculate the Number of Occurrences of Orbit
- ❑ Calculate the Number of Mission Outcome Per Orbit Type
- ❑ Create Landing Outcome Labels
- ❑ Export the Data

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EDA with Data Visualization

To visualize the relationship between Payload Mass and Launch Site or any other variables importance, the following charts were plotted:

- Payload Mass and Launch Site
- Orbit and Outcome
- Payload and Orbit
- Yearly Success Rate
- Flight Number and Orbit
- Flight Number and Launch Site

Create dummy variables to categorical columns

Cast all numeric columns to float64

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EDA with SQL

Perform the following list of SQL to better understand SpaceX dataset:

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA’
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster_versions which have carried the maximum payload mass. Use a subquery
- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

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Build an Interactive Map with Folium

Analyze geospatial data to perform more interactive visual analytics and better understand factors such location and proximity of launch sites that impact launch success rate.

- Mark the location and proximities of launch sites
- Discover patterns via exploring the map faster and more effectively
- Explain how to choose an optimal launch site
- More appalling stories
- Explain why you added those objects

GitHub URL

Build a Dashboard with Plotly Dash

Launch Sites Dropdown List

- Added a dropdown list to enable Launch Site selection.

Pie Chart showing Success Launches (All Sites/Certain Site)

- Added a pie chart to show the total successful launches count for all sites and the Success vs. Failed counts for the site, if a specific Launch Site was selected.

Slider of Payload Mass Range

- Added a slider to select Payload range.

Scatter Chart of Payload Mass vs. Success Rate for the different Booster Versions

- Added a scatter chart to show the correlation between Payload and Launch Success

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Predictive Analysis (Classification)

Analyze geospatial data to perform more interactive visual analytics and better understand factors such location and proximity of launch sites that impact launch success rate.

- Mark the location and proximities of launch sites
- Discover patterns via exploring the map faster and more effectively
- Explain how to choose an optimal launch site
- More appalling stories
- Explain why you added those objects

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Results Summary

Exploratory Data Analysis:

Launch success has improved over time

KSC LC-39A has the highest success rate among landing sites

Orbits ES-L1, GEO, HEO and SSO have a 100% success rate

Visual Analytics:

Most launch sites are near the equator, and all are close to the coast

Launch sites are far enough away from anything a failed launch can damage (city, highway, railway), while still close enough to bring people and material to support launch activities

Predictive Analytics:

Decision Tree model is the best predictive model for the dataset

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Insights Drawn From EDA

Section 2

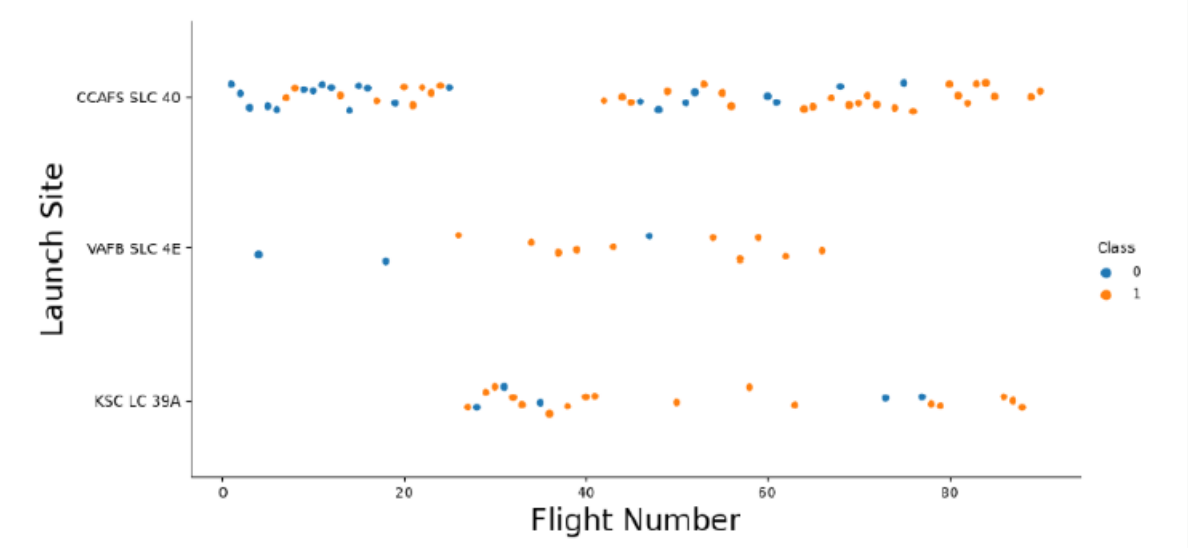
Flight Number vs Launch Site

Explanation:

Earlier flights had a **lower success rate** (**Blue** = **Fail**)

Later flights had a **higher success rate** (**Orange** = **Success**)

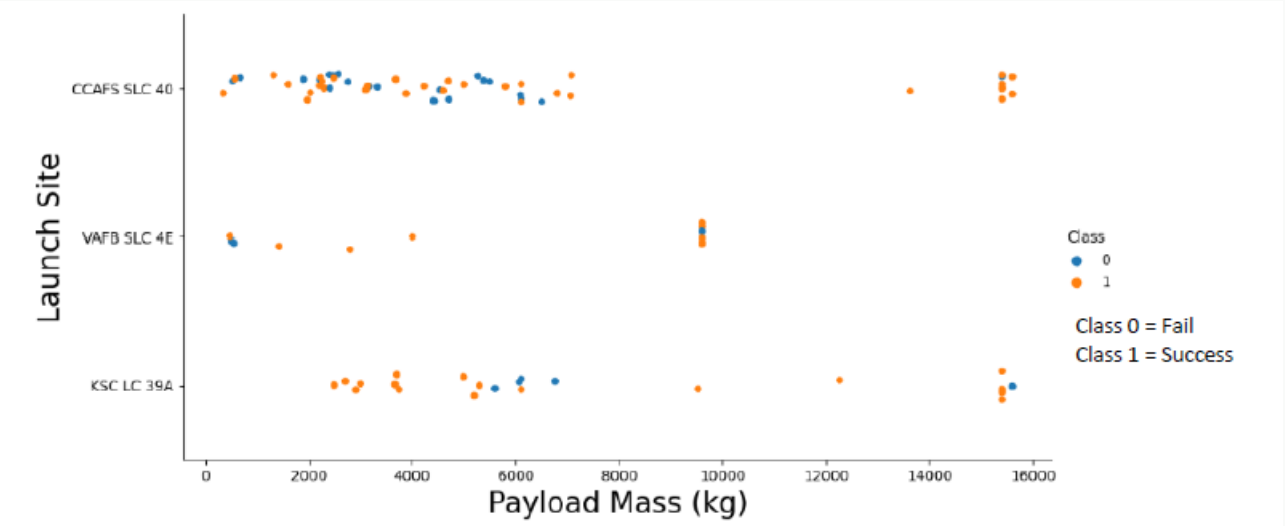
- Around half of launches were from CCAFS SLC 40 launch site
- VAFB SLC 4E and KSC LC 39A have higher success rates
- We can infer that new launches have a higher success rate



Payload vs Launch Site

Explanation:

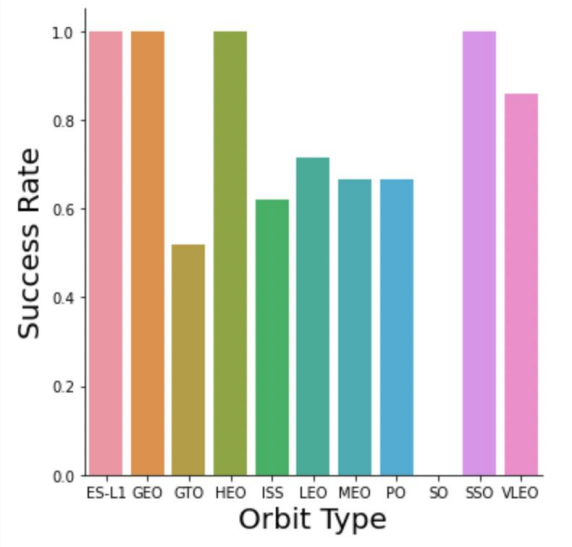
- Typically, the **higher** the **payload mass** (kg), the **higher** the **success rate**
- Most launces with a payload greater than 7,000 kg were successful
- KSC LC 39A has a 100% success rate for launches less than 5,500 kg
- VAFB SKC 4E has not launched anything greater than ~10,000 kg



Success Rate vs Orbit Type

Explanation:

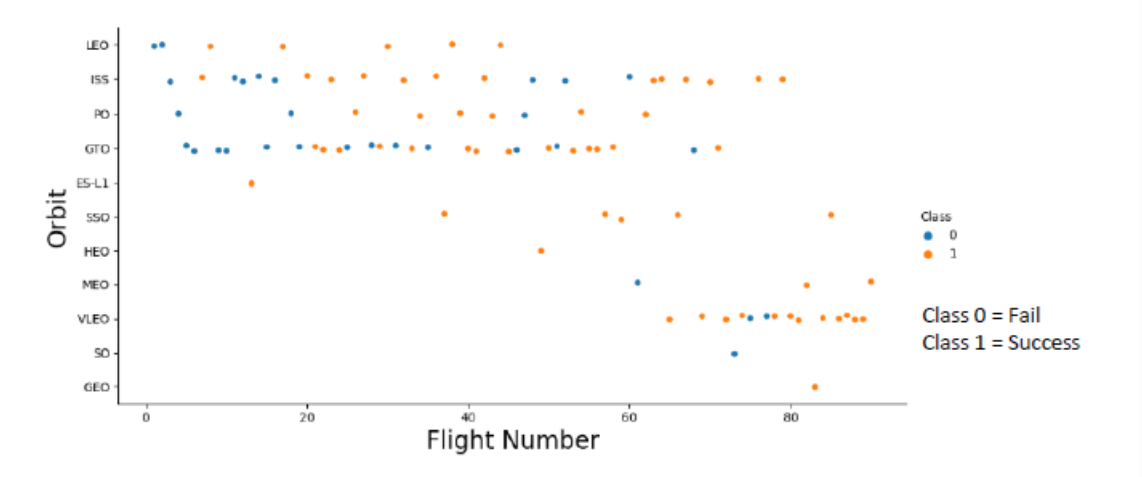
- **100% Success Rate**: ES-L1, GEO, HEO and SSO
- **50%-80% Success Rate**: GTO, ISS, LEO, MEO, PO
- **0% Success Rate**: SO



Flight Number vs Orbit Type

Explanation:

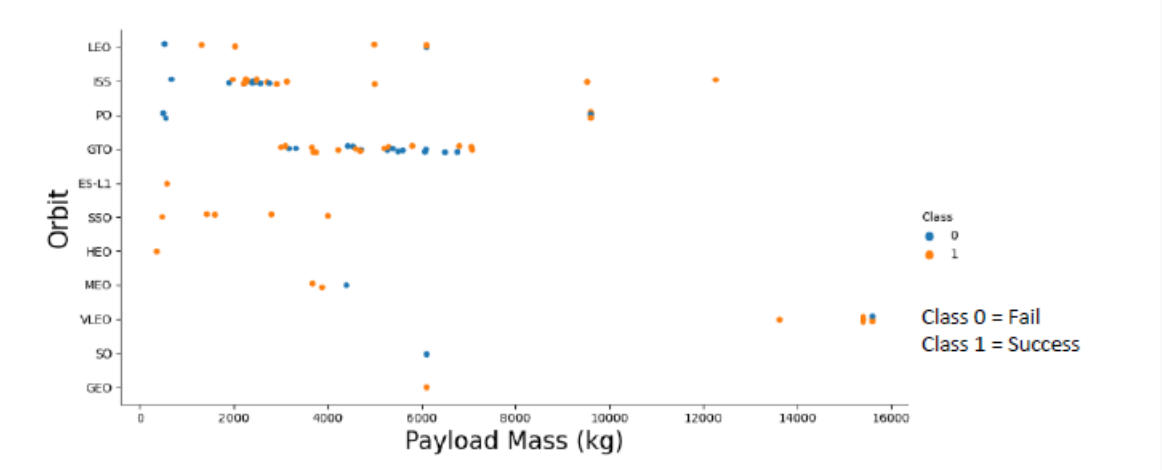
- The success rate typically increases with the number of flights for each orbit
- This relationship is highly apparent for the LEO orbit
- The GTO orbit, however, does not follow this trend



Payload vs Orbit Type

Explanation:

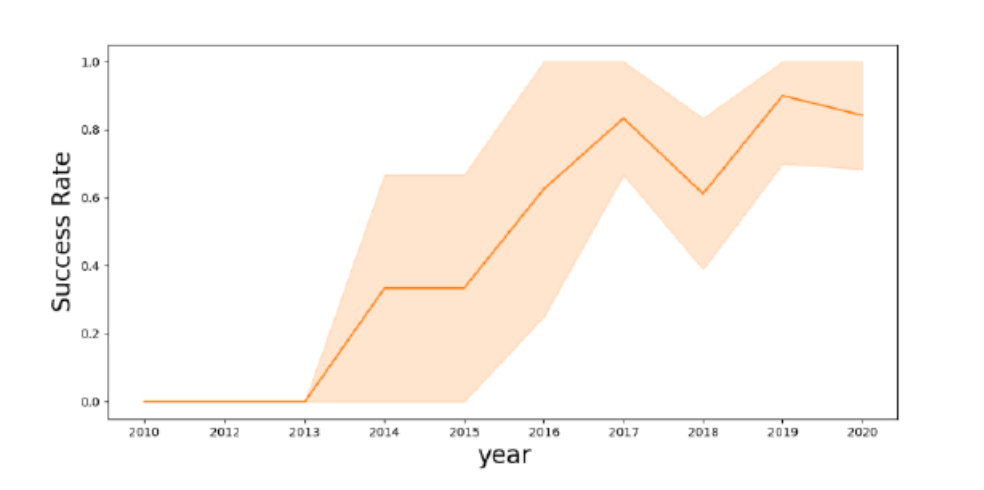
- Heavy payloads are better with LEO, ISS and PO orbits
- The GTO orbit has mixed success with heavier payloads



Launch Success Yearly Trend

Explanation:

- The success rate improved from 2013-2017 and 2018-2019
- The success rate decreased from 2017-2018 and from 2019-2020
- Overall, the success rate has improved since 2013



All Launch Site Names

Launch Site Names:

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

Landing Outcomes:

```
[30]: %sql ibm_db_sa://yyy33800:dwhKg8J3L0IBd6CP@1bbf73c5
      %sql SELECT Unique(LAUNCH_SITE) FROM SPACEXTBL;

 * ibm_db_sa://yyy33800:***@1bbf73c5-d84a-4bb0-85b9
   sqlite:///my_data1.db
Done.
```

[30]:

launch_site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Explanation:
Displaying the names of the unique launch sites in the space mission

Launch Site Names Begin With ‘CCA’

In [5]:
```
%sql select * from SPACEXDATASET where launch_site like 'CCA%' limit 5;
```

```
 * ibm_db_sa://wzf08322:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[5]:

DATE	time__utc_	booster_version	launch_site	payload	payload_mass__kg_	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Explanation:
Displaying 5 records where launch sites begin with the string 'CCA'.

Total Payload Mass

- **45,596 kg** (total) carried by boosters launched by NASA (CRS)

```
In [6]: %sql select sum(payload_mass__kg_) as total_payload_mass from SPACEXDATASET where customer = 'NASA (CRS)';

 * ibm_db_sa://wzf08322:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[6]:

total_payload_mass
45596

Explanation:
Displaying the total payload mass carried by boosters launched by NASA (CRS).

Average Payload Mass F9 v1.1

- **2,928 kg** (average) carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) \
    FROM SPACEXTBL \
    WHERE BOOSTER_VERSION = 'F9 v1.1';
```

```
 * ibm_db_sa://yyy33800:***@1bbf73c5-d84a-4
   sqlite:///my_data1.db
Done.
```

1
2928

Explanation:
Displaying average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

- **12/22/2015**

```
In [8]: %sql select min(date) as first_successful_landing from SPACEXDATASET where landing__outcome = 'Success (ground pad)';

 * ibm_db_sa://wzf08322:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[8]:

first_successful_landing
2015-12-22

Explanation:
Listing the date when the first successful landing outcome in ground pad was achieved

Successful Drone Ship Landing with Payload between 4000 and 6000

- Booster mass greater than 4,000 but less than 6,000
- JSCAT-14, JSCAT-16, SES-10, SES-11 / EchoStar 105

```
%sql SELECT PAYLOAD \
FROM SPACEXTBL \
WHERE LANDING__OUTCOME = 'Success (drone ship)' \
AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
 * ibm_db_sa://yyy33800:***@1bbf73c5-d84a-4bb0-85b9-
   sqlite:///my_data1.db
Done.
```

payload
JCSAT-14
JCSAT-16
SES-10
SES-11 / EchoStar 105

Explanation:
Listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

- 1 Failure in Flight
- 99 Success
- 1 Success (payload status unclear)

```
%sql SELECT MISSION_OUTCOME, COUNT(*) as total_number \
FROM SPACEXTBL \
GROUP BY MISSION_OUTCOME;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation:
Listing the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

```
%sql SELECT BOOSTER_VERSION \
FROM SPACEXTBL \
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Explanation:

Listing the names of the booster versions which have carried the maximum payload mass.

2015 Launch Records

- 1 Failure in Flight
- 99 Success
- 1 Success (payload status unclear)

```
%sql SELECT MISSION_OUTCOME, COUNT(*) as total_number \
FROM SPACEXTBL \
GROUP BY MISSION_OUTCOME;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Explanation:
Listing the failed landing outcomes in drone ship, their booster versions and launch site names for the months in year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Count of Landing Outcomes in descending order

```
%sql SELECT [Landing _Outcome], count(*) as count_outcomes \
FROM SPACEXTBL \
WHERE DATE between '04-06-2010' and '20-03-2017' group by [Landing _Outcome] order by count_outcomes DESC;
```

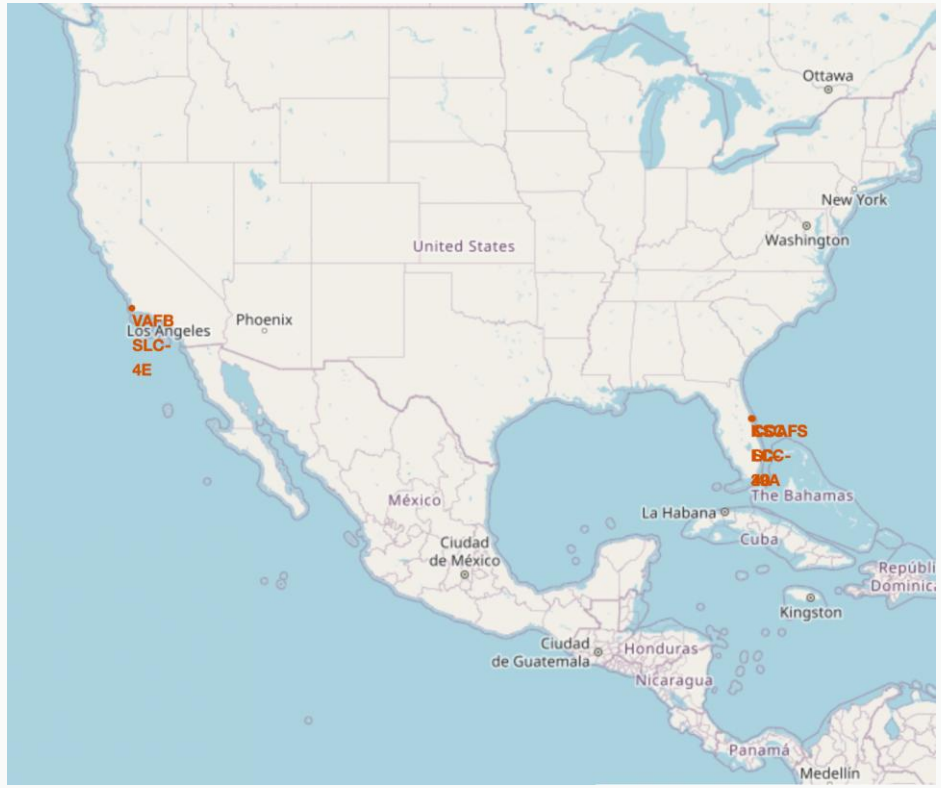
```
 * sqlite:///my_data1.db
Done.
```

Landing _Outcome	count_outcomes
Success	20
No attempt	10
Success (drone ship)	8
Success (ground pad)	6
Failure (drone ship)	4
Failure	3
Controlled (ocean)	3
Failure (parachute)	2
No attempt	1

Explanation:
Ranking the count of landing outcomes Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20 in descending order

LAUNCH SITE PROXIMITIES ANALYSIS

Section 3



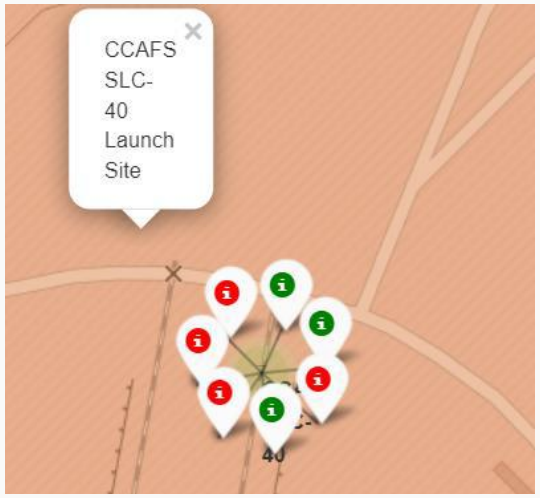
- The closer the launch site to the equator, the easier it is to launch due to equatorial orbit, allowing it to get an additional natural boost.
- Anything on the surface of the Earth at the equator is already moving at 1670 km/hour.
- This in turn, helps to save with the cost of fuel.
- All launch sites are in close proximity to the coast, allowing rockets to be launched towards the ocean.
- The close proximity to the ocean helps in minimizing the risk of potential debris dropping or exploding near people.

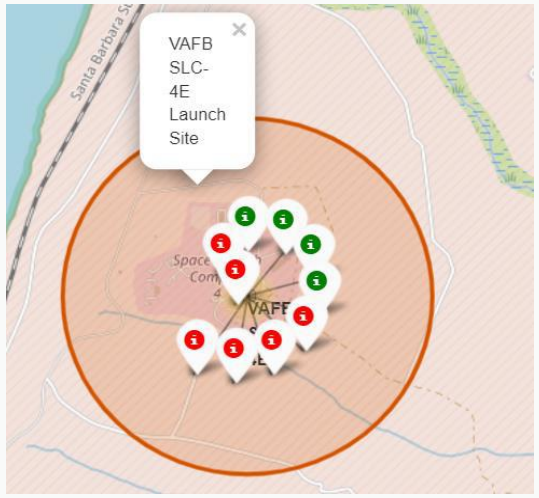
Launch Sites With Markers

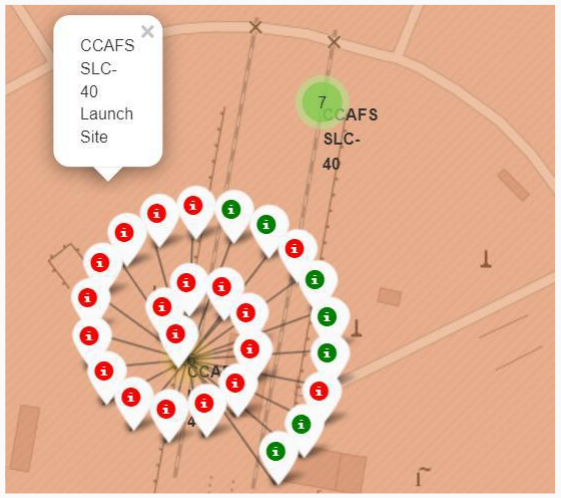
Explanation:

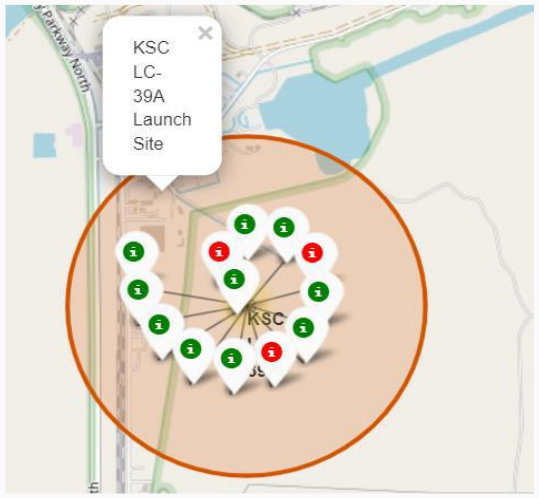
Moving from left to right, it is easy to see the least successful launch site to the most successful launch site based on the color-labeled markers…
Green Marker = Successful Launch - Red Marker = Failed Launch

- Launch Site CCAFS SLC-40 has the lowest success rate.
- Launch Site KSC LC-39A has the highest success Rate.









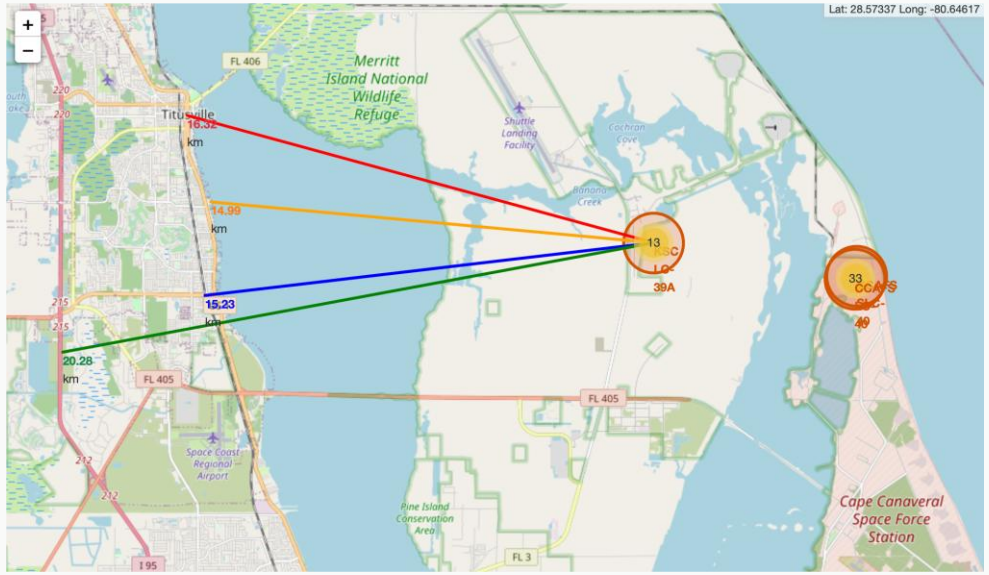
Least Successful → → → → Most Successful

Launch Outcomes With Color Labels

Explanation:

The launch site KSC LC-39A:

- Distance to railway (15.23 km)
- Distance to highway (20.28 km)
- Distance to coastline (14.99 km)
- Distance to closest city Titusville (16.32 km)
- Note that a failed rocket with its high speed can cover distances like 15-20 km in few seconds, which makes it potentially dangerous to populated areas



Distance From Launch Site KSC LC-39A To Proximity Distance Map

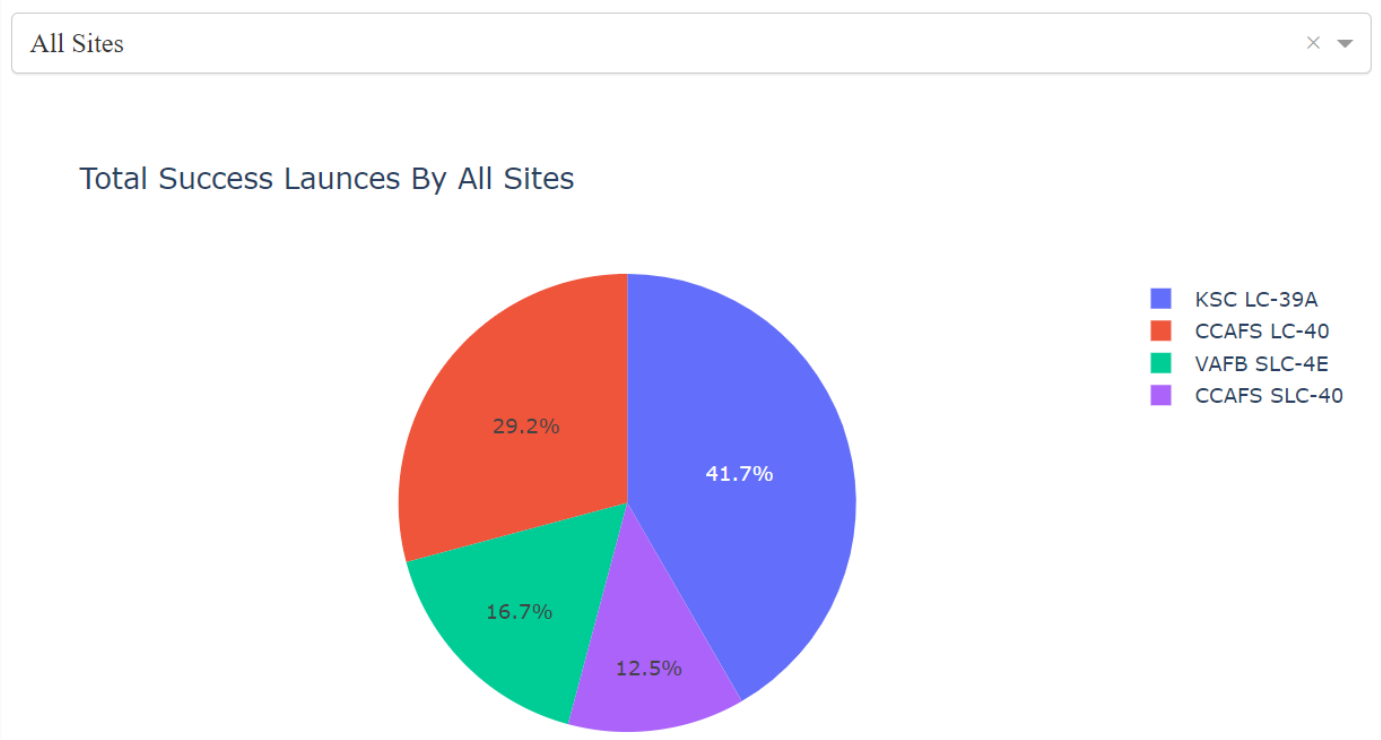
BUILD A DASHBOARD WITH PLOTLY DASH

Section 4

Launch Success Count For All Sites

Explanation:

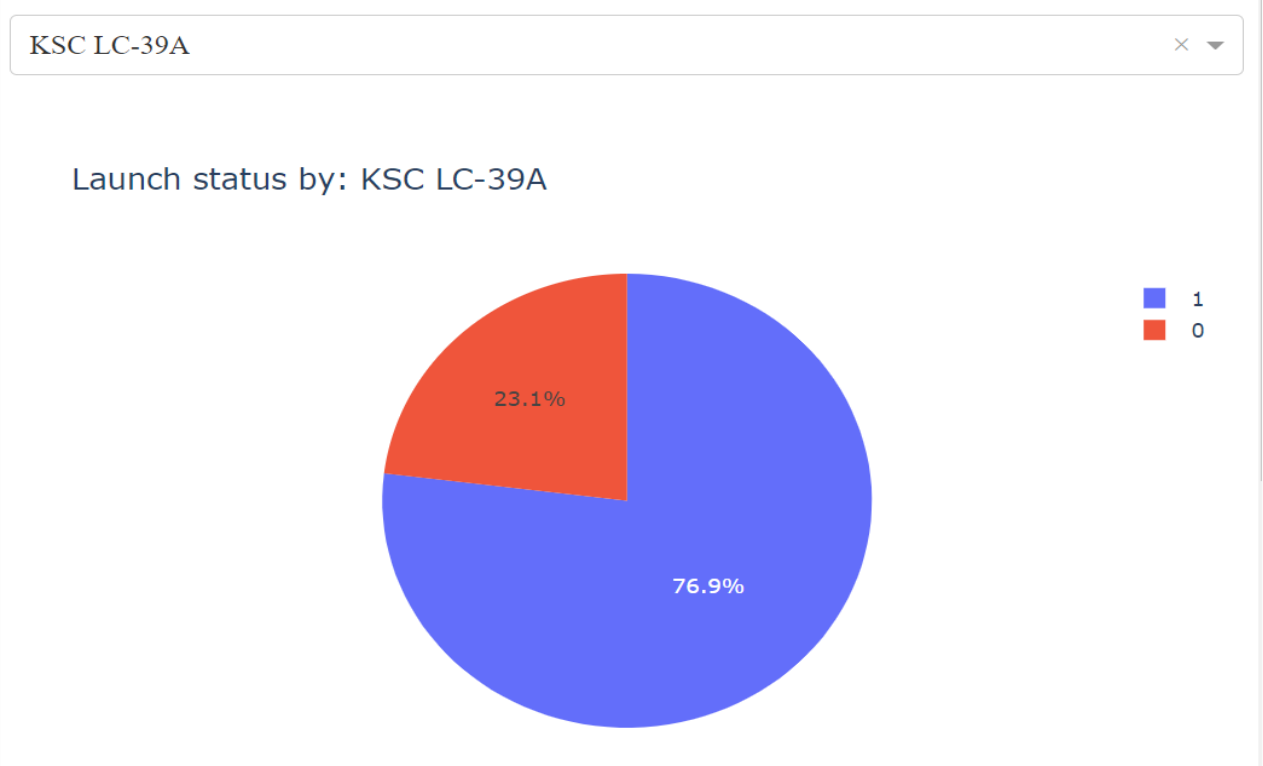
- Launch Site KSC LC-39A has the highest launch success rate
- Launch Site CCAFS SLC-40 has the lowest launch success rate



Launch Site With The Highest Success Rate

Explanation:

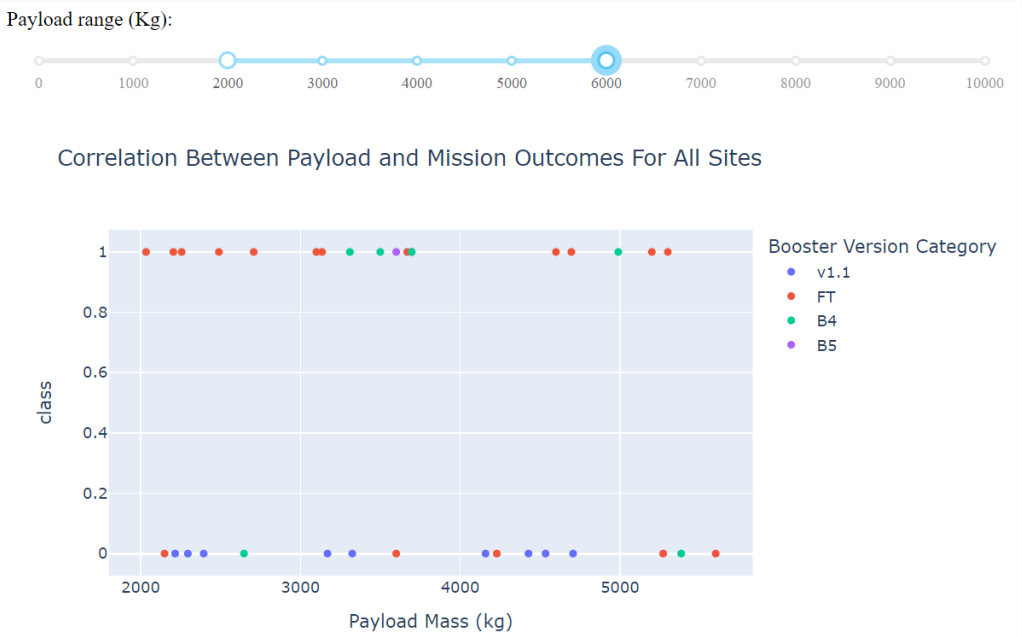
- Launch Site KSC LC-39A has the highest launch success rate
- Launch success rate is 76.9% (10 successes out of 13 attempts)
- Launch failure rate is 23.1% (3 successes out of 13 attempts)

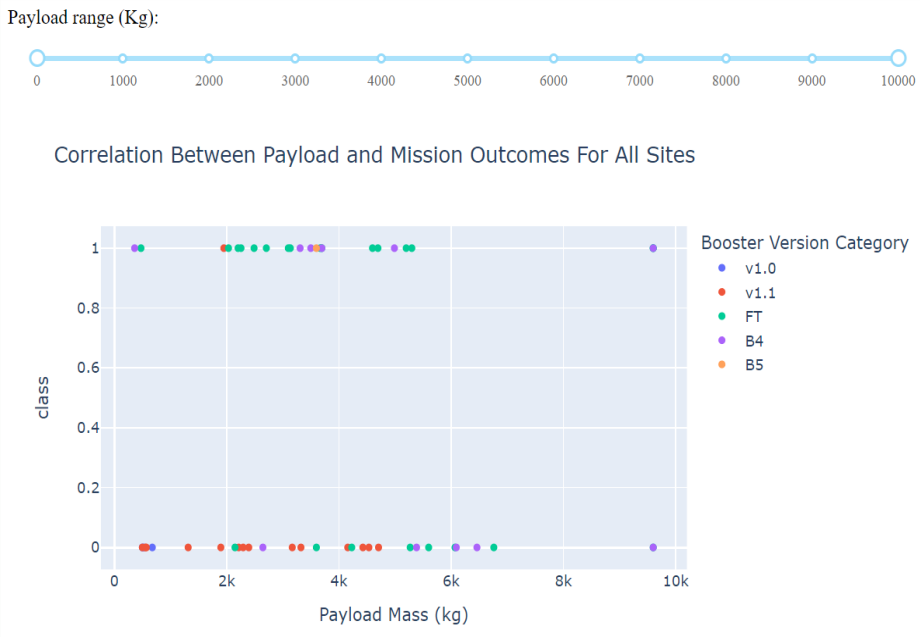


Payload Mass Versus Launch Outcome

Explanation:

- The most successful launches are between 2000 and 5500 KG
- Booster Version Category FT has the most successful launches





PREDICTIVE ANALYSIS (CLASSIFICATION

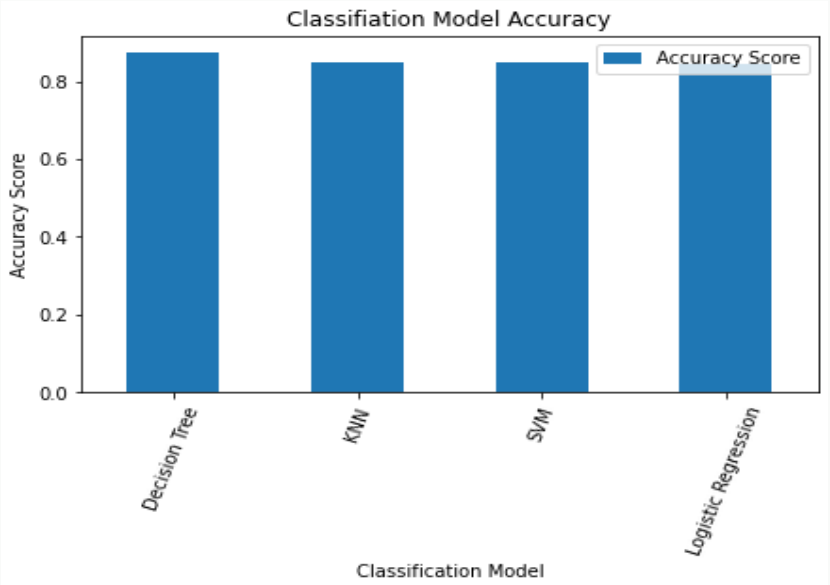
Section 5

Classification Accuracy

Explanation:

- The models all performed at about the same level, with the same test data accuracy scores.
- The same test data accuracy scores may be due to the small test sample size (18 samples). Therefore, we tested all methods based on the whole dataset.
- The best model based on the accuracy scores of the whole dataset tured out to be the Decision Tree (0.875000).

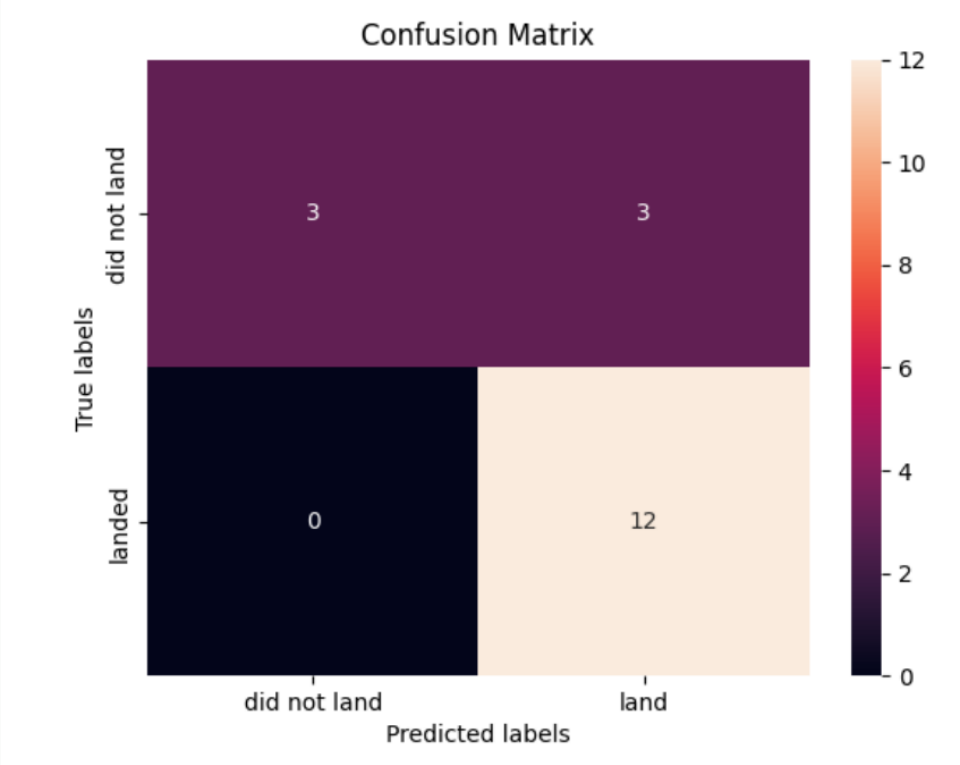
	Algo Type	Accuracy Score	Test Data Accuracy Score
2	Decision Tree	0.875000	0.833333
3	KNN	0.848214	0.833333
1	SVM	0.848214	0.833333
0	Logistic Regression	0.846429	0.833333



Confusion Matrix

Explanation:

- The confusion matrix was the same for all of the models(LR, SVM, Decision Tree, KNN), where the classifier made 18 predictions.
- Confusion Matrix Outputs:
  - 12 True positive (predicted yes = landing)
  - 3 True negative (predicted no = no landing)
  - **3 False positive (predicted yes = no landing)**
- The fact that there are false positives (Type 1 error) is not good
- Overall, the classifier accuracy is about 83%
  - (TP + TN) / (TP + TN + FP + FN) = .833



Conclusions

Model Performance: Decision tree model performed the best, although they all performed similarly on the test

Equator: Most of the launch sites were near the equator for an additional natural boost - due to the rotational speed of earth –which helps save the cost of putting in extra fuel and boosters

Coast: All the launch sites were close to the coast

Launch Success: Increases over time

KSC LC-39A: Had the highest success rate among launch sites and also had a 100% success rate for launches less than 5,500 kg

Orbits: ES-L1, GEO, HEO, and SSO have a 100% success rate

Payload Mass: Across all launch sites, the higher the payload mass (kg), the higher the success rate

THANK YOU